Notes: Vissing-Jørgensen (JPE, 2002)

Limited Asset Market Participation and the Elasticity of Intertemporal Substitution

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1 Big picture

- **Question.** Why have so many estimates of the elasticity of intertemporal substitution (EIS) been close to zero? Does limited asset market participation bias standard Euler-equation estimates downward?
- **Setting.** The paper uses the U.S. Consumer Expenditure Survey (CEX) from 1980:Q1 to 1996:Q1, with the main estimation sample focusing on 1982–1996 because early consumption data are noisier.
- **Core economic idea.** A household's Euler equation for asset i should hold only if the household actually holds asset i at the relevant date. If nonholders are pooled together with holders, the response of expected consumption growth to expected returns is attenuated.
- **Main empirical design.** Estimate log-linearized Euler equations separately for stockholders versus nonstockholders and bondholders versus nonbondholders, using predictable variation in stock and Treasury-bill returns as instruments.
- **Main contribution.** The paper shifts attention from a representative-agent or full-participation view to a participation-based view of the EIS. It argues that the right comparison is not "all households," but rather "households whose portfolio choices make the Euler equation relevant."
- **Main result.** Once households are sorted by whether they hold the relevant asset, the EIS is no longer close to zero. It is around 0.3–0.4 for stockholders and around 0.8–1 for bondholders, while estimates for nonholders are near zero and statistically insignificant.
- **Key heterogeneity result.** The estimated EIS is larger for the richest third of stockholders and the richest third of bondholders.
- **Key robustness result.** The main pattern survives multiple instrument sets, a single-individual-household subsample, and a "median household" approach designed to be more robust to measurement error.
- **Important nuance.** The paper explicitly does *not* interpret near-zero estimates for nonholders as evidence that their true EIS is zero. The point is that the Euler equation for an asset return is misspecified for people who do not hold that asset.
- **Economic takeaway.** Limited participation is a first-order empirical issue in estimating intertemporal substitution. Ignoring it makes households look much less willing to shift consumption across time than they really are.

2 Data and measurement (key objects)

2.1 Why the CEX is useful here

The CEX is useful because it combines repeated consumption interviews with household-level information on financial asset holdings. During the sample period, this lets the paper observe both the *left-hand side* of the Euler equation (consumption growth) and a proxy for whether the household is in the relevant asset market at all.

A second attraction of the setting is that direct stockholding mattered a great deal in the United States, and the CEX explicitly asks about broad asset categories rather than only total financial wealth.

Interpretation. The paper needs a data set that can connect *portfolio participation* to *consumption growth*. The CEX is one of the few household surveys that can do this.

2.2 Sample structure

- The raw CEX covers 1980:Q1 to 1996:Q1.
- Households are interviewed five times at quarterly intervals; the first interview is practice and is not used.
- In each interview, households report consumption for the previous three months.
- Financial asset information is gathered only in interview 5.
- Because the paper needs matched households across interviews 2–5 and drops the noisy early years 1980–1981, the final main sample contains 34,310 semiannual consumption-growth observations.

Interpretation. The unit of observation is not a full household panel over many years. Instead, each household contributes a short but carefully constructed consumption-growth episode that can be linked to asset holdings.

2.3 Defining stockholders and bondholders

The CEX reports holdings in four categories:

- "stocks, bonds, mutual funds, and other such securities,"
- "U.S. savings bonds,"
- "savings accounts," and
- "checking accounts, brokerage accounts, and other similar accounts."

The paper defines:

- **Stockholders** as households with positive holdings in "stocks, bonds, mutual funds, and other such securities."
- **Bondholders** as households with positive holdings in either "stocks, bonds, mutual funds, and other such securities" or "U.S. savings bonds."

Because the Euler equation should hold for households that owned the asset at the *beginning* of the consumption-growth interval, the paper reconstructs beginning-of-period asset status using two extra questions: whether holdings are the same/more/less than a year ago, and by how much dollar holdings changed over the year.

A household is classified as holding the asset at the start of the period if it either:

- reports holding the same amount as a year ago and still holds a positive amount now,

- reports lower holdings than a year ago, or
- reports higher holdings, but by less than its current holdings, so that beginning-of-period holdings must have been positive.

Using this rule, the paper classifies about **21.75%** of households as stockholders and about **31.40%** as bondholders.

Interpretation. The classification is imperfect, but the bias from imperfect classification works against the paper. Misclassifying true holders as nonholders should make the two groups look *more* similar, not less similar.

2.4 Asset-holding layers

To study whether wealthier asset holders behave differently, the paper splits holders into three roughly equal layers based on estimated beginning-of-period real holdings.

For stockholders, the three layers are:

- bottom: \$2 to \$3,587,
- middle: \$3,587 to \$20,264,
- top: above \$20,264,

all in 1982–84 dollars.

For bondholders, the three layers are based on combined holdings of the broad securities category plus U.S. savings bonds:

- bottom: below \$967,
- middle: \$967 to \$9,881,
- top: above \$9,881.

Interpretation. These layers are not just descriptive. They help test whether small asset positions are too noisy or too costly for the Euler equation to work well in practice.

2.5 Consumption measure

The paper constructs consumption as **nondurables plus selected services**, aggregated from detailed CEX categories to match NIPA-style concepts as closely as possible.

It excludes service categories with a large durable component, especially:

- housing expenses (except household operations),
- medical care costs,
- education costs.

Nominal consumption is deflated by the Bureau of Labor Statistics deflator for nondurables for urban households.

Interpretation. The paper wants a consumption concept that is reasonably close to the flow of utility-relevant nondurable consumption, not a measure dominated by lumpy durable spending.

2.6 Semiannual consumption growth and timing

The main consumption-growth object is a six-month growth rate built from adjacent six-month spending blocks:

$$\frac{C_{m+6} + C_{m+7} + C_{m+8} + C_{m+9} + C_{m+10} + C_{m+11}}{C_m + C_{m+1} + C_{m+2} + C_{m+3} + C_{m+4} + C_{m+5}}. \tag{1}$$

The paper prefers **semiannual** to quarterly growth for two reasons:

- households may not reoptimize every quarter, so longer horizons reduce timing mismatch between decisions and survey reporting;
- measurement error is less severe when consumption is aggregated over longer windows.

Extreme outliers in the consumption-growth ratio (below 0.2 or above 5) are dropped. Results are similar if they are retained.

For the relevant asset return over the six-month interval, the paper uses the **middle six months** of returns, roughly $(1+R_{m+2})(1+R_{m+3})\cdots(1+R_{m+7})$, as a simple proxy for the asset return that is most relevant for shifting consumption across the observed interval.

Interpretation. Timing is one of the paper's most practical contributions. The semiannual construction is a response to the fact that households are observed intermittently and may not synchronize consumption decisions to interview dates.

2.7 Why overlapping observations matter

Even though the interview spacing is quarterly, households are interviewed throughout the quarter. This means the data are effectively monthly, but adjacent six-month consumption windows overlap.

As a result, the error term in the log-linearized Euler equation has an **MA(5)** structure when semiannual observations are used.

Interpretation. The serial correlation is mechanical, not behavioral. This is why standard errors must be corrected for both heteroskedasticity and the overlap-induced moving-average error structure.

2.8 Instrument sets and predictive variables

All specifications include 12 seasonal dummies and $\Delta \ln(\text{family size})$. In addition, the paper uses three alternative predictive-instrument sets:

- **Instrument set 1:** log dividend-price ratio.
- **Instrument set 2:** log dividend-price ratio, lagged real stock return, and lagged real Treasury-bill return.
- **Instrument set 3:** log dividend-price ratio, bond horizon premium, and bond default premium.

For semiannual data, the stock-return predictive R^2 is about 0.117, 0.143, and 0.148 across the three sets. For the real Treasury-bill return, the corresponding R^2 values are much higher: about 0.673, 0.674, and 0.744.

Interpretation. The stock-return instrument is much weaker than the Treasury-bill instrument. This fact later helps explain why stock-based EIS estimates are lower and why some overidentification tests behave awkwardly.

2.9 Descriptive facts

Table 1 reports the core descriptive facts:

- average observations per month: about 217 households total, 47 stockholders, 170 nonstockholders, 68 bondholders, and 149 nonbondholders;
- mean semiannual log consumption growth: about 0.014 for stockholders versus −0.001 for nonstockholders;
- mean semiannual log consumption growth: about 0.010 for bondholders versus −0.001 for nonbondholders.

Interpretation. Asset holders are not only wealthier in portfolio terms; they also exhibit higher average consumption growth in the raw data. This already hints that pooling everyone together may blur important heterogeneity.

3 Models and empirical specifications (all equations + interpretation)

3.1 The EIS in certainty and uncertainty

In certainty, the paper defines the EIS as the sensitivity of consumption growth to the relative price of consumption across dates:

$$\text{EIS} = -\frac{d(C_t/C_{t+1})}{d(1+R_{f,t})}\frac{1+R_{f,t}}{C_t/C_{t+1}} = \frac{d\ln(C_{t+1}/C_t)}{d\ln(1+R_{f,t})}. \tag{2}$$

Under uncertainty, if returns and next-period consumption are conditionally jointly lognormal, then for any asset i held in an interior amount,

$$\text{EIS} = \frac{dE_t[\ln(C_{t+1}/C_t)]}{dE_t[\ln(1+R_{i,t})]}. \tag{3}$$

Interpretation. The key conceptual move is that *any* asset with an interior holding can identify the same intertemporal substitution margin. But that statement stops being useful if the household does not hold the asset at all.

3.2 Why limited participation matters for estimation

Let true household consumption be $C_t^{h,*}$ and suppose observed consumption contains multiplicative measurement error:

$$C_t^h = C_t^{h,*} e_t^h. \tag{4}$$

Under CRRA preferences, the true nonlinear Euler equation for household h and asset i is

$$E_t \left[\beta \left(\frac{C_{t+1}^{h,*}}{C_t^{h,*}} \right)^{-\gamma} (1 + R_{i,t}) \right] = 1. \tag{5}$$

Using observed consumption gives

$$E_t \left[\beta \left(\frac{C_{t+1}^{h}}{C_t^{h}} \right)^{-\gamma} (1 + R_{i,t}) \right] = 1. \tag{6}$$

If measurement error is multiplicative and independent of true consumption, returns, and instruments, then measurement error biases the discount-factor estimate but not γ, and hence does not destroy consistency of the EIS estimate.

Interpretation. This is an important technical point. The paper does *not* claim measurement error is harmless in general. It claims that with this particular structure, the parameter governing intertemporal substitution survives, which justifies using the CEX despite noisy household consumption data.

3.3 Cross-sectional aggregation and the "simple cohort" approach

Because each household contributes only one main consumption-growth observation, the paper does not estimate Euler equations household by household. Instead, it aggregates across households of a given type in each period.

For group s (for example, stockholders), the estimating equation is

$$\frac{1}{H_t^s} \sum_{h=1}^{H_t^s} \Delta \ln C_{t+1}^{h,s} = \psi^s \ln(1+R_{s,t}) + \delta_1^s D_1 + \cdots + \delta_{12}^s D_{12} + \alpha^s \frac{1}{H_t^s} \sum_{h=1}^{H_t^s} \Delta \ln(\text{family size})_{t+1}^{h,s} + u_{t+1}^s, \tag{7}$$

where ψ^s is the group-specific EIS estimate.

Interpretation. The paper is effectively estimating one Euler equation for each participation group using a time series of cross-sectional averages. This is the natural way to get around the short panel dimension in the CEX.

3.4 Why family size and seasonality enter the regression

The paper assumes that family size and seasonal factors enter utility multiplicatively. This motivates including:

- monthly seasonal dummies,

- the change in log family size across the consumption interval.

Interpretation. Without these controls, predictable changes in household needs or seasonal consumption patterns could masquerade as intertemporal substitution.

3.5 Estimation method

The main estimator is **linear GMM / IV**, with corrections for:

- heteroskedasticity of arbitrary form,
- MA(5) autocorrelation induced by overlapping six-month windows.

Ordinary least squares is inappropriate because the log asset return is endogenous: the expectational error in returns sits in the regression residual.

For stockholders, the paper also jointly estimates the stock-return and Treasury-bill Euler equations because stockholders are typically also classified as bondholders. Joint estimation exploits cross-equation error correlation and tests whether a common EIS can explain both equations.

Interpretation. The GMM framework is doing two jobs at once: it handles return endogeneity and it lets the paper ask whether the same structural substitution parameter is consistent across different assets.

3.6 Why the paper prefers small instrument sets

The paper emphasizes that adding many weak instruments can push IV estimates toward biased OLS estimates. This is especially important because the stock-return forecasting problem is weak to begin with.

Interpretation. This is why the paper does not throw every macro predictor into the first stage. The goal is not maximal fit of returns, but stable identification of the EIS.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

Identification comes from the idea that predictable movements in asset returns should forecast group-level consumption growth *only* for households that actually hold the asset. The baseline comparison is therefore:

- stockholders versus nonstockholders in the stock Euler equation,
- bondholders versus nonbondholders in the Treasury-bill Euler equation.

If the same return predictor moves expected returns for everyone, but only holders' consumption growth comoves with that return, the difference is informative about limited participation.

Interpretation. The paper is not identifying the EIS from some natural experiment in participation. It is identifying it from a *specification restriction*: the Euler equation should fit holders and fail for nonholders.

4.2 Why reverse causality is weak

At the household level, next-period consumption growth is far too small to move the aggregate stock-market return or Treasury-bill return. The main econometric endogeneity is therefore not reverse causality from household behavior to returns, but expectational error in realized returns.

Interpretation. This is why IV is enough. The problem is not that households set market returns; it is that realized returns are noisy measures of expected returns.

4.3 Measurement error and the median-household check

Even though multiplicative measurement error does not necessarily bias the EIS estimate, the simple-cohort averaging approach may still be fragile when some monthly cross sections are small.

To address this, the paper estimates an alternative "median household" time series: instead of averaging household consumption growth within each group and month, it uses the median observation.

Interpretation. The median-household approach is a robustness check against outliers and small-sample contamination in the cross section. The fact that the results get even stronger under this approach is important.

4.4 Why the single-individual-household subsample matters

The paper repeats the analysis using households with only one individual in both periods. This has two advantages:

- family-size adjustment becomes less important,
- the household's intertemporal decision problem is simpler.

Interpretation. If the results only appeared in large, complicated households, one might suspect mismeasurement of family structure or intrahousehold aggregation. The stronger estimates for single-individual households work against this concern.

4.5 Joint stock–bond estimation: what works and what does not

When the stock and Treasury-bill Euler equations are jointly estimated for stockholders with a common EIS, the overidentification tests reject for stockholders as well as nonstockholders.

The paper interprets this as consistent with the much weaker predictive power of stock-return instruments. In other words, the stock equation may still be too noisy to line up cleanly with the Treasury-bill equation even for actual asset holders.

Interpretation. This is one of the paper's most honest features. It does not claim every specification works beautifully. The cleanest support for the theory comes from the separate estimations and the median-household robustness, not from the joint overidentification tests.

4.6 What the paper can and cannot distinguish

The paper can convincingly show that pooling nonholders with holders biases EIS estimates downward. But it cannot fully distinguish among all deeper reasons why nonholders look different.

In particular:

- it cannot perfectly classify stock and bond market participation because the CEX asset categories are coarse;
- it cannot fully separate true heterogeneity in EIS from differences in transaction costs, diversification, or optimization sophistication across wealth groups;
- it does not estimate risk aversion, so it does not by itself resolve the equity premium puzzle.

Interpretation. The strongest claim is narrow and important: *limited participation matters for EIS estimation.* The paper is not claiming to solve every consumption-based asset-pricing problem at once.

4.7 Taxes and savings-account holders

A further nuance is that the paper uses pretax returns because accurate household-specific capital-income tax rates are not available in the CEX. Ignoring taxes likely biases EIS estimates downward, since after-tax returns fluctuate less than pretax returns.

The paper also checks households that hold savings accounts but not stocks, bonds, mutual funds, or U.S. savings bonds. Their estimated EIS is small and unstable, which the author interprets as consistent with either tiny financial wealth or effective borrowing/borrowing-constraint status.

Interpretation. These last two points both strengthen the paper's main message. Taxes would make the true EIS larger, and the failure of savings-account-only households to fit the Treasury-bill Euler equation suggests that "nominal asset holding" is not the same as meaningful intertemporal optimization margin.

5 Tables

5.1 Table 1: summary statistics and why the holder/nonholder split matters

Read:

- Table 1 reports average monthly cross-sectional counts and average semiannual log consumption growth for each household group.
- Stockholders and bondholders are much smaller groups than nonholders, but they have noticeably higher mean consumption growth.
- The top stockholder and top bondholder layers are especially small groups, which helps explain why their standard errors are large even when the point estimates are economically big.

TABLE 1
SUMMARY STATISTICS, CEX DATA, 1982–96

Group	Mean Number of Observations per Month	Mean of $(1/H_t)\sum_{h=1}^{H_t}\Delta\ln C_{t+1}^h$ over the Sample	Standard Deviation of $(1/H_t)\sum_{h=1}^{H_t}\Delta\ln C_{t+1}^h$ over the Sample
	A. Semiannual Data, All Household Sizes		
All	217	.002	.024
Stockholders	47	.014	.041
Nonstockholders	170	−.001	.028
Bottom stockholder layer	11	.022	.088
Middle stockholder layer	11	.002	.101
Top stockholder layer	11	.014	.102
Bondholders	68	.010	.034
Nonbondholders	149	−.001	.030
Bottom bondholder layer	17	.005	.069
Middle bondholder layer	17	.012	.072
Top bondholder layer	17	.008	.073
	B. Semiannual Data, Single-Individual Households		
All	51	−.001	.045
Stockholders	10	.009	.098
Nonstockholders	41	−.003	.052
Bondholders	13	.002	.085
Nonbondholders	39	−.002	.054

NOTE.—The variable $(1/H_t)\sum_{h=1}^{H_t}\Delta\ln C_{t+1}$ is seasonally adjusted using seasonal dummies. The seasonally adjusted value is the sample mean of the series plus the residual from a regression on 12 dummies.

Economic message. The table makes the empirical challenge concrete: the households for whom the stock Euler equation should hold are a minority of the sample. If one simply pools all households, the identifying signal is mechanically diluted.

5.2 Table 2: separate Euler-equation estimations are the core result

TABLE 2
GMM ESTIMATION OF LOG-LINEARIZED EULER EQUATIONS: REAL TREASURY BILL RETURN AND REAL VALUE-WEIGHTED NYSE RETURN, SEPARATE ESTIMATIONS (CEX, 1982–96, Semiannual Data)

	Instrument Set 1		Instrument Set 2			Instrument Set 3		
	$\hat{\sigma}$	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ
	A. Euler Equation for Stocks 1. All Household Sizes							
All	.098		.066	.086		.068	.314	
	(.071)		(.062)			(.059)		
Stockholders	**.299**		.281	.260		.200	.018	
	(.146)		(.114)			(.091)		
Nonstockholders	**.057**		.017	.048		.049	.568	
	(.079)		(.070)			(.070)		
Bottom layer	.046		−.054	.570		.052	.828	
	(.186)		(.163)			(.158)		
Middle layer	.175		.350	.547		.173	.027	
	(.274)		(.207)			(.261)		
Top layer	.486		.417	.203		.292	.027	
	(.325)		(.235)			(.188)		
Nonstockholders vs. stockholders		3.255			4.340			3.296
		(.071)			(.037)			(.069)
Nonstockholders vs. top layer		2.146			2.378			2.941
		(.143)			(.123)			(.086)
	2. Single-Individual Households							
All	.202		−.008	.025		.261	.439	
	(.172)		(.106)			(.139)		
Stockholders	.698		.323	.005		.681	.969	
	(.496)		(.264)			(.326)		

TABLE 2
(Continued)

	Instrument Set 1		Instrument Set 2			Instrument Set 3		
	$\hat{\sigma}$	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ
Nonstockholders	.077		−.115	.071		.160	.250	
	(.143)		(.137)			(.123)		
Nonstockholders vs. stockholders		1.733			1.958			3.866
		(.188)			(.162)			(.049)
	B. Euler Equation for Treasury Bills 1. All Household Sizes							
All	.372		.362	.097		.264	.143	
	(.232)		(.225)			(.220)		
Bondholders	**.932**		.839	.215		.783	.147	
	(.368)		(.360)			(.353)		
Nonbondholders	**.105**		.087	.024		.005	.382	
	(.270)		(.257)			(.244)		
Bottom layer	.986		.798	.492		.726	.244	
	(.662)		(.612)			(.580)		
Middle layer	.287		.336	.476		.277	.627	
	(.550)		(.544)			(.523)		
Top layer	1.648		1.672	.356		1.530	.067	
	(.515)		(.505)			(.505)		
Nonbondholders vs. bondholders		4.029			3.192			4.127
		(.045)			(.074)			(.042)
Nonbondholders vs. top layer		8.055			8.905			7.871
		(.005)			(.003)			(.005)
	2. Single-Individual Households							
All	.681		.524	.032		.282	.053	
	(.456)		(.450)			(.411)		

Bondholders	2.624		2.759	.163		1.617	.007	
	(.885)		(.880)			(.824)		
Nonbondholders	.048		−.028	.039		−.105	.103	
	(.479)		(.475)			(.460)		
Nonbondholders vs. bondholders		8.371			7.527			4.722
		(.004)			(.006)			(.030)

NOTE.—Numbers in parentheses are standard errors for $\hat{\sigma}$ and p-values for the Wald test. For the overidentification test, the entries are p-values, and the test has two degrees of freedom for each of instrument sets 2 and 3. Twelve monthly dummies are included as explanatory variables and instruments. The estimations for all household sizes furthermore include $\Delta\ln$ (family size) as an explanatory variable and instrument. In addition the instrument sets include the following variables. Instrument set 1: log dividend-price ratio. Instrument set 2: log dividend-price ratio, lagged log real value-weighted NYSE return, and lagged log real Treasury bill return. Instrument set 3: log dividend-price ratio, default premium, and bond horizon premium.

Read:

- For the stock-return Euler equation, the all-households estimate is only about 0.10 in the baseline specification, but the stockholder estimate rises to about 0.30, while the nonstockholder estimate stays near 0.06.
- Within stockholders, the top asset-holding layer has the largest estimate, around 0.49 in the baseline and still around 0.29–0.42 in the richer instrument sets.
- For the Treasury-bill Euler equation, the all-households estimate is about 0.37, the bondholder estimate is about 0.93, and the nonbondholder estimate is about 0.11 in the baseline specification.
- The top bondholder layer has the biggest EIS estimate, around 1.65 in the baseline and still around 1.53–1.67 with the richer instrument sets.
- The same qualitative pattern appears for the single-individual-household subsample, often with even bigger point estimates, though standard errors are wider.

Economic message. Table 2 is the heart of the paper. The same return predictors that imply a near-zero EIS in pooled data imply a clearly positive EIS once the sample is restricted to households for whom the Euler equation is actually relevant.

5.3 Table 3: joint stock–bond estimation is informative but imperfect

TABLE 3
GMM Estimation of Log-Linearized Euler Equations: Real Treasury Bill Return and Real Value-Weighted NYSE Return, Joint Estimations (CEX, 1982–96, Semiannual Data)

	Instrument Set 1			Instrument Set 2			Instrument Set 3		
	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ
	A. All Household Sizes								
All	.080 (.077)	.025		.046 (.058)	.036		.026 (.045)	.016	
Stockholders	**.442 (.144)**	.022		.404 (.114)	.071		.261 (.079)	.011	
Nonstockholders	**.013 (.057)**	.036		.003 (.022)	.041		.003 (.033)	.028	
Bottom layer	.006 (.054)	.141		.003 (.021)	.372		.007 (.026)	.078	
Middle layer	.068 (.173)	.175		.242 (.203)	.218		.048 (.040)	.041	
Top layer	.700 (.354)	.020		.422 (.260)	.190		.324 (.179)	.004	
Nonstockholders vs. stockholders			9.228 (.002)			12.886 (.0003)			13.677 (.0002)
Nonstockholders vs. top layer of stockholders			3.890 (.049)			2.127 (.145)			4.211 (.040)
	B. Single-Individual Households								
All	.187 (.192)	.055		.024 (.041)	.036		.141 (.120)	.062	
Stockholders	1.108 (.527)	.028		.378 (.306)	.023		.568 (.278)	.043	
Nonstockholders	.018 (.089)	.114		−.028 (.050)	.068		.050 (.074)	.079	
Nonstockholders vs. stockholders			4.161* (.041)			1.716* (.190)			3.250* (.071)

Note.—Numbers in parentheses are standard errors for $\hat{\sigma}$ and p-values for the Wald test. For the overidentification test, the entries are p-values, and the test has one degree of freedom for instrument set 1 and five degrees of freedom for instrument sets 2 and 3. Twelve monthly dummies are included as explanatory variables and instruments. The estimations for all household sizes furthermore include $\Delta \ln$ (family size) as an explanatory variable and instrument. In addition the instrument sets include the following variables. Instrument set 1: log dividend-price ratio. Instrument set 2: log dividend-price ratio, lagged log real value-weighted NYSE return, and lagged log real Treasury bill return. Instrument set 3: log dividend-price ratio, default premium, and bond horizon premium.

* These Wald tests do not allow for correlation of the error terms across the Euler equations for stocks and Treasury bills. If this were allowed, one of the matrices involved would not be positive definite.

Read:

- Table 3 jointly estimates the stock and Treasury-bill Euler equations for stockholders and imposes a common EIS across the two equations.
- The common-EIS estimate for stockholders is roughly 0.44, 0.40, and 0.26 across the three instrument sets. For nonstockholders it remains near zero.
- However, the overidentification tests reject for stockholders as well as for nonstockholders.

Economic message. The table is useful mainly as a warning. The participation story is strong, but the stock-return equation is still noisy enough that imposing one common EIS across stock and bond Euler equations creates tension even within the holder sample.

5.4 Table 4: the median-household approach makes the main pattern even stronger

TABLE 4
GMM Estimation of Log-Linearized Euler Equations: Real Treasury Bill Return and Real Value-Weighted NYSE Return, Separate Estimations (CEX, 1982–96, Semiannual Data, Median Household Approach)

	Instrument Set 1		Instrument Set 2			Instrument Set 3		
	$\hat{\sigma}$	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ
	A. Euler Equation for Stocks — 1. All Household Sizes							
All	.125 (.081)		.069 (.067)	.016		.138 (.064)	.691	
Stockholders	**.437 (.229)**		.451 (.175)	.147		.287 (.105)	.014	
Nonstockholders	**.055 (.081)**		−.014 (.073)	.033		.113 (.074)	.352	
Bottom layer	.054 (.231)		.082 (.191)	.320		−.016 (.201)	.408	
Middle layer	−.082 (.340)		.067 (.233)	.097		.228 (.278)	.006	
Top layer	1.072 (.798)		.933 (.538)	.019		.706 (.367)	.450	
Nonstockholders vs. stockholders		2.833 (.092)			4.900 (.027)			4.203 (.040)
Nonstockholders vs. top layer		1.687 (.194)			2.331 (.127)			4.116 (.042)
	2. Single-Individual Households							
All	.348 (.257)		.168 (.145)	.028		.347 (.182)	.877	
Stockholders	1.152 (.897)		1.030 (.622)	.090		.874 (.479)	.661	
Nonstockholders	.238 (.206)		.044 (.127)	.058		.268 (.153)	.624	
Nonstockholders vs. stockholders		1.350 (.245)			3.853 (.050)			3.438 (.064)
	B. Euler Equation for Treasury Bills — 1. All Household Sizes							
All	.420 (.270)		.413 (.262)	.057		.231 (.263)	.010	
Bondholders	**1.380 (.507)**		1.140 (.472)	.189		.870 (.425)	.047	
Nonbondholders	**−.011 (.290)**		−.101 (.283)	.052		−.168 (.268)	.153	
Bottom layer	1.345 (.712)		1.162 (.677)	.060		1.214 (.591)	.700	
Middle layer	.267 (.691)		.255 (.689)	.295		.182 (.660)	.075	
Top layer	2.838 (.585)		2.877 (.565)	.346		2.390 (.575)	.017	
Nonbondholders vs. bondholders		7.476 (.006)			8.733 (.003)			7.077 (.008)
Nonbondholders vs. top layer		22.514 (.000)			24.930 (.000)			20.094 (.000)
	2. Single-Individual Households							
All	1.174 (.504)		1.018 (.490)	.155		.661 (.433)	.073	

TABLE 4
(*Continued*)

	Instrument Set 1		Instrument Set 2			Instrument Set 3		
	$\hat{\sigma}$	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ	$\hat{\sigma}$	Overidentification Test	Wald Test Equals σ
Bondholders	2.625 (.801)		2.727 (.794)	.253		2.256 (.726)	.032	
Nonbondholders	.740 (.617)		.589 (.610)	.085		.379 (.564)	.047	
Nonbondholders vs. bondholders		3.788 (.052)			4.326 (.038)			4.187 (.041)

Note.—Numbers in parentheses are standard errors for $\hat{\sigma}$ and p-values for the Wald test. For the overidentification test, the entries are p-values, and the test has two degrees of freedom for each of instrument sets 2 and 3.. Twelve monthly dummies are included as explanatory variables and instruments. The estimations for all household sizes furthermore include $\Delta \ln$ (family size) as an explanatory variable and instrument. In addition the instrument sets include the following variables. Instrument set 1: log dividend-price ratio. Instrument set 2: log dividend-price ratio, lagged log real value-weighted NYSE return, and lagged log real Treasury bill return. Instrument set 3: log dividend-price ratio, default premium, and bond horizon premium.

Read:

- Table 4 redoes the main estimations using the median rather than the mean household within each month-group cell.

- For stockholders, the EIS rises to around 0.44–0.45 under the first two instrument sets, compared to about 0.30 in the baseline table.
- For bondholders, the EIS rises to roughly 1.14–1.38 under the first two instrument sets, compared to roughly 0.93 in the baseline table.
- The top bondholder layer reaches very large estimates, around 2.3–2.9, while nonbondholder estimates remain much smaller.

Economic message. Table 4 is the paper's strongest robustness exercise. If anything, the simple cohort means were muting the limited-participation effect rather than generating it.

6 Short summary

- The paper asks whether ignoring limited asset market participation biases estimates of the elasticity of intertemporal substitution downward.
- It uses the U.S. Consumer Expenditure Survey and estimates separate log-linearized Euler equations for stockholders versus nonstockholders and bondholders versus nonbondholders.
- The main identifying idea is that the Euler equation for asset i should hold only for households with an interior position in asset i.
- Stockholders have EIS estimates around 0.3–0.4; bondholders have estimates around 0.8–1; richer asset-holder layers have even larger estimates.
- Nonstockholders and nonbondholders have estimates near zero, which the paper interprets as misspecification rather than evidence of a truly zero EIS.
- The results survive multiple instrument sets, single-individual-household estimation, and a median-household robustness check.
- The joint overidentification tests are less supportive, which the paper attributes to the weakness of stock-return instruments relative to Treasury-bill instruments.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that limited asset market participation is a major reason why standard Euler-equation estimates of the EIS can look implausibly small. Once estimation is restricted to households that actually hold the relevant asset, the EIS is clearly positive and economically meaningful.

7.2 Economic intuition

The economic intuition is simple but powerful. A household that does not own stocks has no reason to adjust its consumption growth in response to predictable stock-market returns. If such households are mixed together with stockholders, the average response of consumption growth to returns gets flattened. The same logic applies to bonds and Treasury bills.

7.3 Takeaway

The paper's takeaway is methodological: before estimating an Euler equation, one must ask whether the households in the sample are actually on the margin that the equation describes. For the EIS, participation is not a side issue—it is part of the identification problem itself.